# Regime-Switching Models

Chapter 10

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# Outline

- Why parameters switch
- The two-state Markov-switching model
- The Hamilton filter
- Estimation and empirical results
- Threshold models: A brief tour
- MS-VAR: A brief map

# Why parameters switch

Section 1

# Why parameters switch

- So far, time-series models assume constant parameters
- Convenient but not obviously true, as we saw with structural breaks
- GDP example
  - A single equation estimated between 1955 and 2009 is confounding
    - Different monetary regimes (Gold exchange standard vs fiat money)
    - Different regulatory and fiscal policies (policy changes after 2009 financial crisis)
    - Different international environment (Vietnam War, the Cold War, etc.)
    - Structural shocks (9/11)
    - Etc…

# Why parameters switch

- The structural break tests **describe** and tell us about potential parameter changes.
- The structural break tests are the first step toward building a regime-switching model

# The two-state Markov-switching model

Section 2

# The two-state Markov-switching model

- Two-state first-order Markov chain
  - Assume two states $(s_t)$: Expansion and recession
  - $s_t \in \{0,1\}$
  - The chain is memory-less; tomorrow's regime depends only on today's, not on the full history
  - $P = \begin{bmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{bmatrix}, \quad p_{ij} = \Pr(s_t = j | s_{t-1} = i)$
  - Each row adds to 1: $p_{00} + p_{01} = 1; p_{10} + p_{11} = 1$
  - This produces only two free parameters; which are estimated from the data
  - If $p_{00} \approx 1 \rightarrow$ Expansion persists
  - If $p_{11} \approx 2 \rightarrow$ Recession persists

# The two-state Markov-switching model

- Regime duration
  - Assume $p_{00} = 0.94$ and that you are looking at quarterly data
  - Staying at state $i$ for $k$ periods requires $k - 1$ successive "stay" draws, followed by one "leave"
    - $p_{ii}^{k-1}(1 - p_{ii})$
    - Expected regime-duration: $\mathbb{E}[s_i] = \frac{1}{1-p_{ii}} = \frac{1}{1-0.94} = \frac{1}{0.06} \approx 16.7$ quarters

# The two-state Markov-switching model

- For a long sample, what fraction of quarters does the economy spend in each regime?
  - $\pi_0 = \frac{1-p_{11}}{(1-p_{00})+(1-p_{11})}, \quad \pi_0 + \pi_1 = 1$
  - Assume $(p_{00}, p_{11}) = (0.94, 0.75)$
  - $\pi_0 = \frac{1-0.75}{(1-0.94)+(1-0.75)} = \frac{0.25}{0.06+0.25} = \frac{0.25}{0.31} \approx 0.81$

# The two-state Markov-switching model

- The MS-AR(k) Model
  - Markov-switching AR of order $k$
    - $y_t = \bar{y}_{s_t} + \sum_{j=1}^{k} \phi_j \left( y_{t-j} - \bar{\bar{y}}_{s_{t-j}} \right) + \varepsilon_t, \quad \varepsilon_t \sim N\left(0, \sigma_{s_t}^2\right)$
    - Full parameter list
      - $\bar{y}_0, \bar{y}_1, \sigma_0^2, \sigma_1^2, \phi_1, \ldots, \phi_k, p_{00}, p_{11}$

| Specification | Mean Switches | Variance switches | Extra parameters |
|---|---|---|---|
| Mean-switching | Yes | No | $\bar{y}_0, \bar{y}_1, \sigma^2$ |
| Variance-switching | No | Yes | $\bar{y}, \sigma_1^2, \sigma_2^2$ |
| Both | Yes | Yes | $\bar{y}_0, \bar{y}_1, \sigma_0^2, \sigma_1^2$ |

# The two-state Markov-switching model

- Why is this just not an ARMA model with dummies?
  - The probability-transition matrix $P$ assumes that history doesn't matter
  - But… history does matter
  - Conditioning on **today's** regime is not enough
  - Regime-correction depends on the regime observed at **every** past lag
  - The Hamilton filter allows to deal with this complication

# The Hamilton filter

Section 3

# The Hamilton filter

- Given everything observed today, what's the probability the economy is currently in each regime?
  - Markov
    - $P = \begin{bmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{bmatrix}, \quad p_{ij} = \Pr(s_t = j | s_{t-1} = i)$
  - Hamilton
    - $H = \begin{bmatrix} h_{00} & h_{01} \\ h_{10} & h_{11} \end{bmatrix}, \quad h_{ij} = \Pr(s_t = j | y_1, \dots, y_t)$
  - Problem
    - $s_t$ is hidden, it needs to be filtered out from observed $\{y_t\}$
    - The Hamilton filter extracts $s_t$ from $\{y_t\}$ (observations)

# The Hamilton filter

- Each date has two probabilities
  - Predicted probability: $\xi_{t|t-1}$ (before observing $y_t$) → one-step regime forecast
  - Filtered probabilities: $\xi_{t|t}$ (using data through $y_t$)

# The Hamilton filter

- Step 1: Prediction
  - $\xi_{t|t-1} = H'\xi_{t-1|t-1}$
  - Written in linear form:
    - $\Pr(s_t|\mathcal{F}_{t-1}) = p_{00}\Pr(s_{t-1}|\mathcal{F}_{t-1}) + p_{01}\Pr(s_{t-1}|\mathcal{F}_{t-1})$
  - Written in Spanglish
    - The predicted probability of expansion tomorrow is the “probability of expansion today and staying” plus “probability of recession today and switching”
    - Note that $y_t$ is not used yet — this is a projection with everything know on $t-1$ ($\mathcal{F}_{t-1}$)

# The Hamilton filter

- Step 2: Update
  - When $y_t$ arrives, update $\xi_{t|t-1}$ using Bayes' rule
  - Let $y_t \sim N(\bar{y}, \sigma_y)$
  - $\Pr(y_t|s_t = j, \mathcal{F}_{t-1}) = \frac{1}{\sqrt{2\pi\sigma_j^2}} e^{\left(-\frac{(y_t-\bar{y}_j)^2}{2\sigma_j^2}\right)}$

- $\Pr(s_t = j|\mathcal{F}_t) = \frac{\Pr(y_t|s_t = j, \mathcal{F}_{t-1}) \cdot \Pr(s_t = j|\mathcal{F}_t)}{\underbrace{\sum_m \Pr(y_t|s_t = m, \mathcal{F}_{t-1}) \cdot \Pr(s_t = m|\mathcal{F}_t)}_{1}}$

# The Hamilton filter

- Example $(t = 1)$
  - $(\bar{y}_0, \bar{y}_1) = (3, -2)$
  - $\sigma^2 = 4$
  - $(p_{00}, p_{11}) = (0.9, 0.7)$
  - $(\pi_0, \pi_1) = (0.75, 0.25)$
  - $y_1 = 4.2$ — a strong expansion quarter, $4.2 > 3$
    - $\Pr(y_1 | s_1 = 0) = \phi\left(\frac{4.2-3}{2}\right) = \phi(0.60) \approx 0.333$
    - $\Pr(y_1 | s_1 = 1) = \phi\left(\frac{4.2+2}{2}\right) = \phi(3.10) \approx 0.003$
    - $\Pr(s_1 | y_1) = \frac{0.333 \times 0.75}{(0.333 \times 0.75) + (0.003 \times 0.25)} = \frac{0.250}{00.251} \approx 0.996$
    - One quarter of **strong growth** pushes the probability of being in expansion to 99.6%

# The Hamilton filter

- Example $(t = 2)$
  - Predict forward state first, then $y_2 = -3.8$ arrives (a sharp contraction)
  - $\Pr(s_2|y_1) = 0.90(0.996) + 0.30(0.004) \approx 0.897$
  - Expansion is high persistence; but with $y_2 = -3.8$:
    - $\Pr(y_2|s_2 = 0) = \phi(-3.40) \approx 0.001$
    - $\Pr(y_2|s_2 = 1) = \phi(-0.90) \approx 0.266$
    - $\Pr(s_2|y_1, y_2) = \frac{0.001 \times 0.897}{(0.001 \times 0.897) + (0.266 \times 0.103)} \approx 0.032$
    - One quarter of **sharp contraction** drives the expansion probability from 99.6% to 3.2%

# The Hamilton filter

- Filtered vs smoothed probabilities
  - Filtered $(\xi_{t|t})$: real-time, uses data through $t$ (what forecaster would have used **at the time**)
  - Smoothed $(\xi_{t|T})$: uses the **entire** sample, the filter is run forward and backwards

# Estimation and empirical results

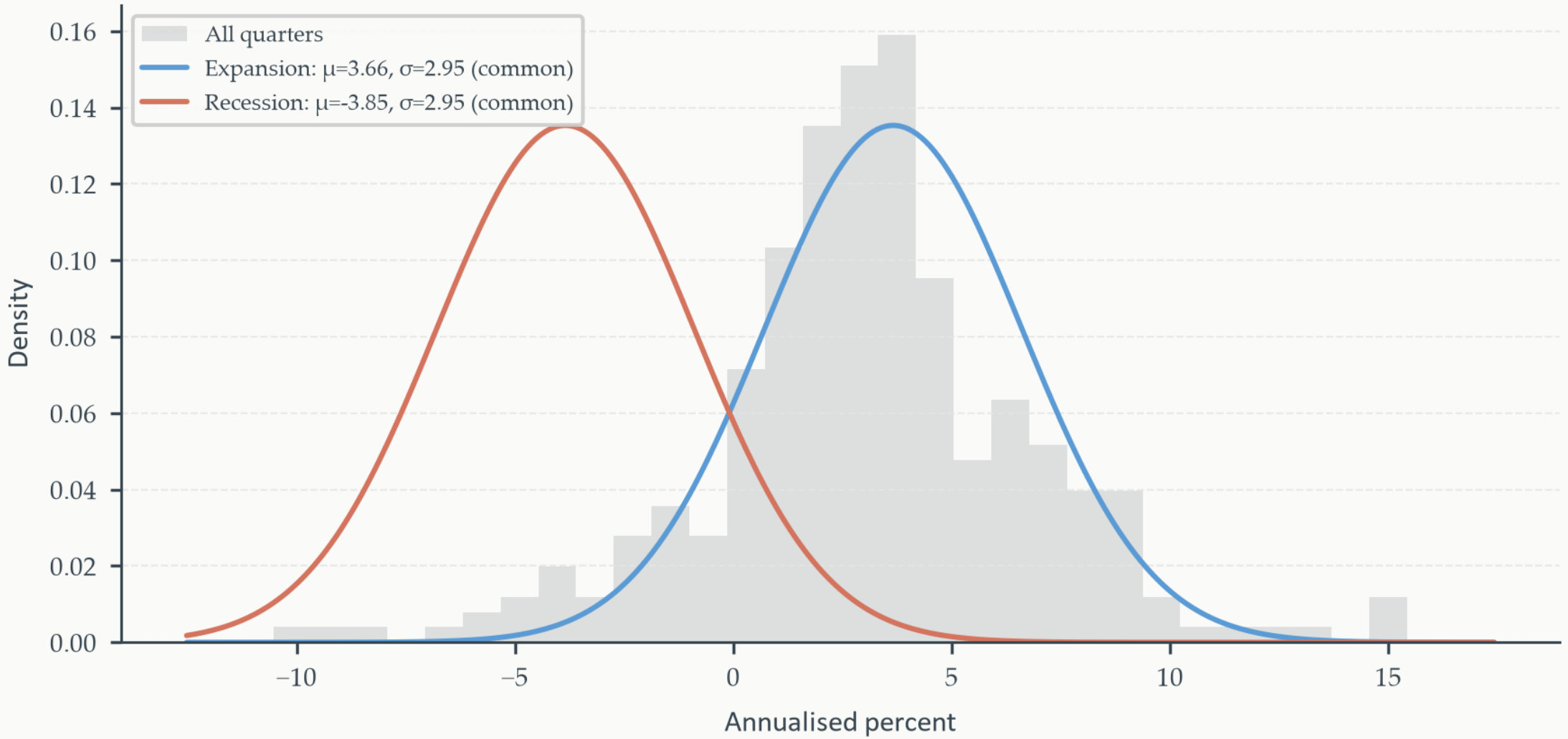
Section 4

# Estimation and empirical results

- Maximum likelihood (via the filter)
  - $\max_{\theta} \ell(\theta) = \sum_{t=1}^{T} \log \Pr(y_t | \mathcal{F}_{t-1}; \theta)$
  - $\theta = (\bar{y}_0, \bar{y}_1, \sigma_0^2, \sigma_1^2, \phi_0, \phi_1, p_{00}, p_{11})$

| Parameter | Expansion | Conctraction |
|---|---|---|
| $\hat{\bar{y}}$ | 3.66** | -3.85** |
| $\hat{\sigma}$ | 2.95 | 2.95 |
| $p_{ii}$ | 0.956 | 0.400 |

## Estimated Regime Distributions — MS-AR(2)



# Threshold models: A brief tour

Section 5

# Thresholds models: A brief tour

- Two ways to build a regime-switching model
  - Markov-switching
    - $s_t$ is unobserved and governed by transition probabilities
    - $s_t$ must be inferred from observed data
    - Fits better when the driving force is really latent
  - Threshold models
    - $s_t$ es determined by whether an **observable** variable $q_t$ sits above or below a threshold $\gamma$
    - Fits better when $q_t$ serves as the trigger of each regime

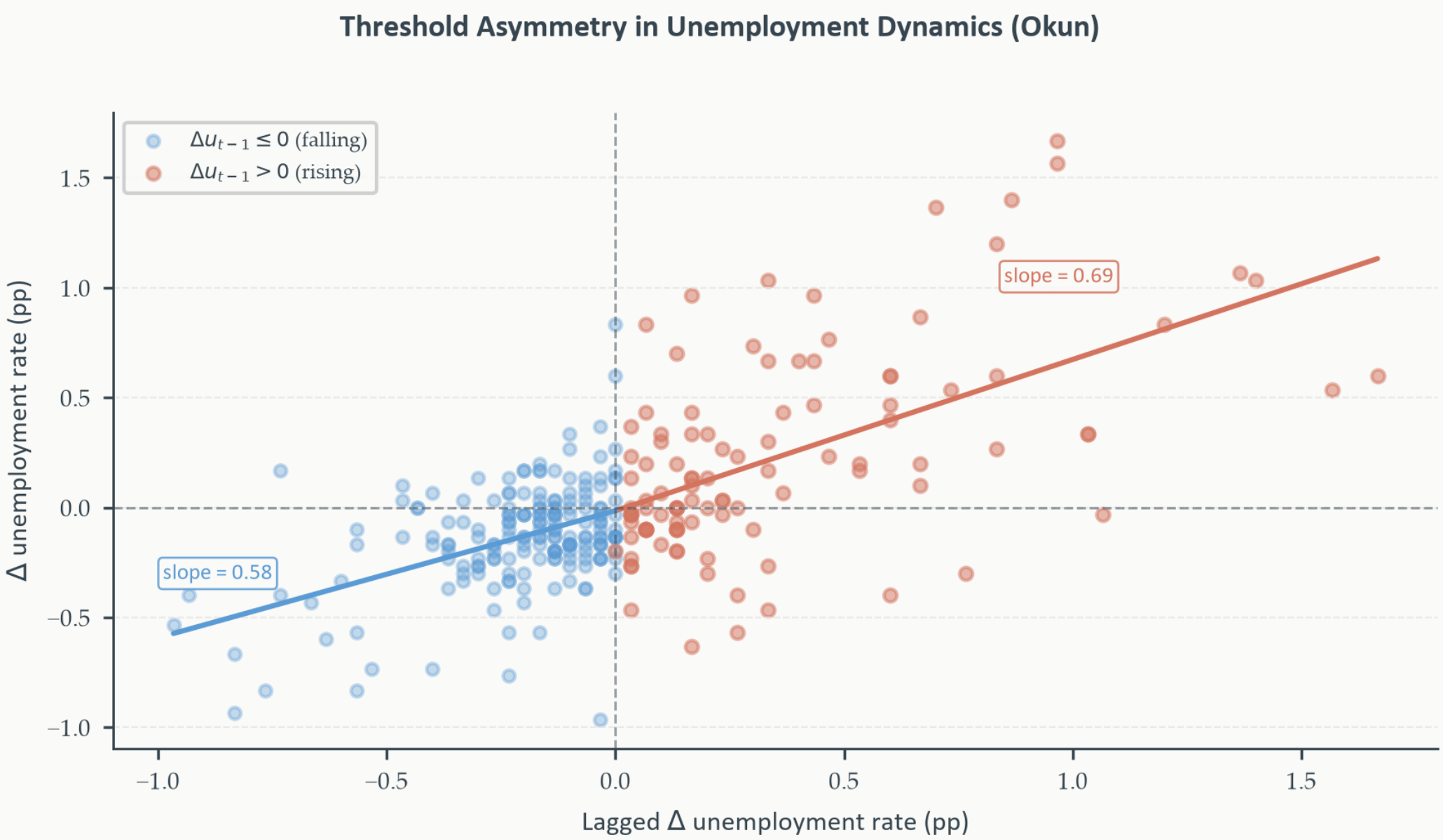
# Thresholds models: A brief tour

- TAR (Threshold Autoregression) models
  - AR coefficients switch depending on whether $y_{t-d}$ crosses threshold $\gamma$
  - $y_t = \begin{cases} \phi_{1,0} + \phi_{1,1} y_{t-1} + \cdots + \phi_{1,p} y_{t-p} + \varepsilon_t, & y_{t-d} \leq \gamma \\ \phi_{2,0} + \phi_{2,1} y_{t-1} + \cdots + \phi_{2,p} y_{t-p} + \varepsilon_t, & y_{t-d} > \gamma \end{cases}$

  - If the threshold variable is lag of itself $(q_t = y_{t-d})$, then this is a Self-Exciting TAR (SETAR) model
  - Do GPD dynamics change depending on whether the last quarter's growth was negative or positive?

# Thresholds models: A brief tour

- SETAR estimation
  - Conditional OLS
    - For any candidate $\gamma$, split the sample by $y_{t-d} \leq \gamma$ or not
    - Estimate two separate AR regressions by OLS
    - Collect the SSR
    - Find the $\hat{\gamma}$ that minimizes total SSR across all $\gamma$ candidates


Threshold Asymmetry in Unemployment Dynamics (Okun)
$\Delta u_{t-1} \leq 0$ (falling)
$\Delta u_{t-1} > 0$ (rising)
slope = 0.58
slope = 0.69
Δ unemployment rate (pp)
Lagged Δ unemployment rate (pp)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
1.5

# Thresholds models: A brief tour

- Smooth Transitions
  - TAR (and SETAR) assumes a sharp change of regime
  - What if the transition itself is gradual?
  - Fit a Smooth Transition Autoregression (STAR) model
    - Let $G_t$ be a smooth function: $G_t = G(q_t, \gamma, c) \in (0,1)$
    - $y_t = (1 - G_t)(\phi_{0,0} + \phi_{0,1} y_{t-1} + \cdots + \phi_{0,p} y_{t-p}) + G_t(\phi_{1,0} + \phi_{1,1} y_{t-1} + \cdots + \phi_{1,p} y_{t-p}) + \varepsilon_t$

    - Where:
      - $q_t$: transition variable
      - $c$: location
      - $\gamma$: speed

# Thresholds models: A brief tour

- Two smoothing functions
  - Logistic
    - LSTAR = $G_t = \frac{1}{1+e^{-\gamma(q_t-c)}}$
  - Exponential
    - ESTAR = $G_t = 1 - e^{-\gamma(q_t-c)^2}$

# MS-VAR: A brief map

Section 6

# MS-VAR: A brief map

- The MS-VAR model expands the approach of the univariate regime-switch models to a VAR model
- A MS-VAR can model different IRFs depending on the active regime

# MS-VAR: A brief map

- A landscape of models — which can also be combined
  - MSI — Intercept switching
  - MSM — Mean switching
  - MSA — AR coefficient switching
  - MSH — Heteroskedasticity switching

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